

A Three-Stage Skeleton-Based Framework for Vietnamese Sign Language Recognition Under Long-Tail Vocabulary: SignGloss-Align, ProtoSign, and CurriVSL

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Abstract—Vietnamese Sign Language (VSL) recognition faces severe data scarcity and long-tail vocabulary: a large proportion of sign classes have only one or few samples. We propose a three-stage skeleton-based framework comprising three named models: SignGloss-Align (SGA)—contrastive sign-text pretraining with optional masked-pose reconstruction; ProtoSign (PS)—few-shot Prototypical Network trained in episodic N-way K-shot fashion; and CurriVSL (CVSL)—curriculum-based class-balanced fine-tuning with a cosine classifier. All three share a common temporal skeleton backbone (temporal convolution, BiLSTM, attention pooling) on MediaPipe Holistic keypoints. Experiments on the VSL dataset [1] show that CurriVSL achieves 98.6% Top-1 accuracy on a held-out test set over the full vocabulary (3,782 classes), outperforming SGA (88.77%) and ProtoSign (89.08%), an RGB baseline R3D-18 (95.3%), and substantially outperforming skeleton baselines (LSTM 83.4%, BiLSTM 72.6%, 1D-CNN 71.8%, attention 78.2%, transformer 52.0%). We provide a detailed comparison of the three models (training dynamics, sample efficiency, and sensitivity to long-tail classes) and discuss challenges specific to Vietnamese sign language data.

Index Terms—Vietnamese Sign Language, sign language recognition, SignGloss-Align (SGA), ProtoSign, CurriVSL, skeleton-based recognition, contrastive learning, few-shot learning, curriculum learning, MediaPipe Holistic, long-tail distribution.

I. INTRODUCTION

Sign language recognition (SLR) aims to map continuous or isolated sign videos to glosses or spoken-language text. In this work we focus on *isolated* recognition: the system receives **one video of a single sign action** as input and produces **one vocabulary item (gloss)** as output—i.e., the recognized sign in the form of its lexical label (e.g., the Vietnamese gloss for that sign). The full pipeline is: (1) preprocess the input video to extract a skeleton sequence (e.g., MediaPipe Holistic keypoints); (2) feed the skeleton sequence through the trained model (backbone + classification head); (3) take the argmax of the output logits to obtain the predicted class index; (4) map the class index to the corresponding gloss (vocabulary). Thus, *input: one sign-action video; output: the recognized vocabulary (gloss)*. For under-resourced languages such as Vietnamese Sign Language (VSL), the main obstacles are limited labeled data and a long-tail vocabulary: many

sign classes have only one or a few samples [1]. Standard supervised training with cross-entropy on such data leads to poor generalization and underfitting on rare classes.

We address these challenges with a *three-stage skeleton-based framework* that reuses a single shared backbone across stages. We introduce three named models, each corresponding to one stage:

- **SignGloss-Align (SGA)**: Pretrains the backbone with sign-text contrastive learning (InfoNCE) and optional masked-pose reconstruction, aligning skeleton sequences with gloss text embeddings. SGA does not use a classification head at pretraining time.
- **ProtoSign (PS)**: Trains a Prototypical Network head in an episodic N-way K-shot setting on top of the (optionally SGA-pretrained) backbone, so the model learns from limited support examples per class.
- **CurriVSL (CVSL)**: Fine-tunes the full model with a curriculum over class-frequency tiers and class-balanced sampling, using a cosine similarity classifier. CVSL is the full pipeline (backbone + curriculum + cosine head) and is the main proposed model for deployment.

All three operate on the same precomputed skeleton sequences (MediaPipe Holistic, with **110 hand+face joints** as the primary setup and 42 hand joints as ablation), ensuring consistency and reproducibility.

Our contributions are: (1) a unified three-stage pipeline for VSL with *named* models (SGA, ProtoSign, CurriVSL) that explicitly target long-tail and few-sample classes; (2) rigorous comparison of SGA, ProtoSign, and CurriVSL on a common test set, including training dynamics and sample efficiency; (3) ablation against skeleton baselines (LSTM, BiLSTM, 1D-CNN, attention pooling, transformer); (4) in-depth analysis of Vietnamese data challenges and the role of each stage. We build on the VSL dataset and evaluation protocol from [1], which provides 6,046 original videos over 3,782 unique signs from 11 signers, with stratified splitting to guarantee vocabulary coverage.

II. RELATED WORK

A. Vietnamese Sign Language and the VSL Dataset

Vietnamese Sign Language has been formally studied since the 1990s, with regional variants (Hanoi, Hai Phong, Ho Chi Minh City) and ongoing standardization. Prior VSL datasets were small (50–500 signs, 1–3 signers) and often captured in controlled settings [1]. The VSL dataset [1] derives from the QIPEDC (Quality Improvement of Primary Education for Deaf Children) project: 6,046 original videos, 3,782 unique signs, 11 signers, with stratified 80%/10%/10% train/validation/test splitting *before* augmentation to avoid leakage. After augmentation (5 variants per training video), the training set grows to 29,022 samples. A critical difficulty is class imbalance: 64% of classes have only one sample, and 33.4% have 2–5 samples. The authors report baseline accuracies from 42.18% (LSTM) to 58.92% (Video Swin Transformer) under a vocabulary-coverage-first protocol and emphasize the need for few-shot and transfer learning for single-sample classes.

B. Skeleton-Based and Multimodal SLR

Skeleton-based SLR uses 2D/3D keypoints (hands, body, face) instead of raw RGB to reduce variability and computation. MediaPipe Holistic provides 21 hand landmarks per hand (42 total), 33 pose points, and 468 facial landmarks; the latter capture non-manual markers important for sign grammar [1]. Graph convolutional networks (e.g., ST-GCN), temporal convolutions, LSTMs, and transformers are commonly used on skeleton sequences. Our backbone combines temporal convolution blocks, BiLSTM, and attention pooling, and we keep the same 42 hand-joint input for fair comparison with the VSL paper’s hand-only and hand+pose setups where applicable.

C. Contrastive and Few-Shot Learning for SLR

Contrastive pretraining aligns visual (or skeleton) representations with text (gloss) in a shared space via losses such as InfoNCE, improving downstream recognition and zero-shot capability. Few-shot SLR uses episodic training (N-way K-shot) and metric learning (e.g., Prototypical Networks) so that models can generalize from few support examples. We combine both: SGA gives a gloss-aligned backbone; ProtoSign trains a ProtoNet head on the same backbone; CurriVSL then fine-tunes on the full vocabulary with a curriculum and class-balanced sampling to handle the long tail.

III. DATASET AND CHALLENGES

A. VSL Dataset

We use the Vietnamese Sign Language (VSL) dataset presented in [1]. It comprises:

- **6,046** original video recordings from the QIPEDC project.
- **3,782** unique Vietnamese sign language gestures (vocabulary size).
- **11** signers with diverse signing styles.

- Stratified split: 80% train, 10% validation, 10% test on *original* videos before augmentation; augmentation applied only to the training set.

B. Key Challenges for Vietnamese Data

- 1) **Long-tail distribution:** 64% of classes have only one sample; 33.4% have 2–5 samples; only 2.6% have 6 or more. This makes standard cross-entropy training biased toward frequent classes and underperforming on rare ones.
- 2) **Vocabulary coverage:** The primary goal of SLR is to recognize a comprehensive vocabulary. Evaluation must ensure all test classes appear in training (vocabulary-coverage-first protocol [1]).
- 3) **Non-manual markers:** Facial landmarks (e.g., 468 MediaPipe points) carry grammatical information. In this version, our primary setup already uses **hand+face (110 joints)**; the 42-joint hand-only setup is retained for ablation.
- 4) **Signer and style diversity:** With 11 signers, inter-signer variation is non-negligible; augmentation and robust temporal modeling are essential.

C. Preprocessing and Keypoint Extraction

The end-to-end flow is: **input one sign-action video** → extract skeleton → model → **output one vocabulary (gloss)**. Videos are decoded and sampled to a fixed number of frames (e.g., 16). We extract 2D/3D hand keypoints using **MediaPipe Holistic**: 21 landmarks per hand (42 total), (x, y, z) per landmark. Keypoints are normalized and missing detections can be interpolated from adjacent frames. Skeleton sequences are stored as arrays of shape (T, V, C) with T frames, $V = 42$ joints, $C = 3$ coordinates, and aligned (padding or subsampling) to a fixed T for batching. At inference, one skeleton (from one video) is fed to the model; the output logits are argmaxed to a class index, which is mapped to the gloss (vocabulary). Figure 1 illustrates the keypoint extraction pipeline.

IV. METHODOLOGY

A. Overview of the Three-Stage Framework and Named Models

The pipeline shares one **skeleton backbone** across all stages. The backbone maps a skeleton sequence $\mathbf{X} \in \mathbb{R}^{T \times V \times C}$ to a video embedding $\mathbf{z} \in \mathbb{R}^D$ (and optionally per-frame features for masked-pose). The three *named* models differ in training objective and head:

- **SignGloss-Align (SGA):** Pretrain backbone + projection layers with sign–text contrastive (InfoNCE) and masked-pose reconstruction; no classification head. At evaluation time a linear head can be trained on top of the frozen backbone.
- **ProtoSign (PS):** Fix or fine-tune backbone; train a Prototypical Network head with episodic N-way K-shot and cross-entropy on distance-based logits. At evaluation time

a linear head can be trained on the frozen backbone for standard classification.

- **CurriVSL (CVSL):** Fine-tune backbone + cosine classifier with curriculum (high $\rightarrow$ high+mid $\rightarrow$ all classes) and class-balanced sampling. This is the full proposed model for final recognition.

B. Shared Skeleton Backbone

The backbone accepts input $\mathbf{X} \in \mathbb{R}^{B \times T \times V \times C}$ and produces $\mathbf{z} \in \mathbb{R}^{B \times D}$.

1) *Temporal Convolution Blocks:* Two blocks of 2D convolution over time and joints: kernel $(k, 1)$ so that the filter slides along the time dimension. Each block: Conv2D $\rightarrow$ BatchNorm $\rightarrow$ GELU $\rightarrow$ Dropout. Output of the second block has shape $[B, T, V, C']$ with $C' = 128$.

2) *Sequence Modeling and Pooling:* The tensor is reshaped to $[B, T, V \cdot C']$ and passed through LayerNorm, then a 2-layer bidirectional LSTM (hidden size 256, so output dimension $D = 512$). Finally, a single learnable query $\mathbf{q} \in \mathbb{R}^{1 \times 1 \times D}$ is used for attention over the LSTM output:

$$\alpha_t = \frac{\exp(\mathbf{q}^\top \mathbf{h}_t / \sqrt{D})}{\sum_{t'} \exp(\mathbf{q}^\top \mathbf{h}_{t'} / \sqrt{D})}, \quad \mathbf{z} = \sum_t \alpha_t \mathbf{h}_t. \quad (1)$$

So the backbone outputs a single vector $\mathbf{z}$ per video (and optionally the sequence $\{\mathbf{h}_t\}$ for masked-pose).

C. SignGloss-Align (SGA): Sign-Text Contrastive and Masked-Pose Pretraining

Let $\mathbf{z}$ be the backbone output and $\mathbf{e}$ the gloss text embedding (e.g., from a frozen sentence encoder). We project both to a common dimension P and compute InfoNCE.

1) *Projections:* $\mathbf{v} = \text{norm}(\mathbf{W}_v \mathbf{z})$, $\mathbf{t} = \text{norm}(\mathbf{W}_t \mathbf{e})$, with $\mathbf{W}_v \in \mathbb{R}^{P \times D}$, $\mathbf{W}_t \in \mathbb{R}^{P \times D_e}$.

2) *InfoNCE Loss:* For a batch of B pairs (z_i, e_i) :

$$\mathcal{L}_{\text{cont}} = -\frac{1}{2B} \sum_{i=1}^B \left(\log \frac{\exp(\mathbf{v}_i^\top \mathbf{t}_i / \tau)}{\sum_{j=1}^B \exp(\mathbf{v}_i^\top \mathbf{t}_j / \tau)} + \log \frac{\exp(\mathbf{t}_i^\top \mathbf{v}_i / \tau)}{\sum_{j=1}^B \exp(\mathbf{t}_j^\top \mathbf{v}_i / \tau)} \right) \quad (2)$$

where τ is the temperature (e.g., 0.07).

3) *Masked-Pose Reconstruction (Optional):* A subset of frames is masked; the model predicts the original joint coordinates from the backbone’s per-frame features using an MLP. Loss is MSE on masked positions only, encouraging the backbone to retain pose information.

D. ProtoSign (PS): Few-Shot Prototypical Head

In each episode we sample N classes and K support examples per class, plus Q query examples per class. Let $\mathcal{S}_c$ be the set of support indices for class c . The prototype for class c is

$$\mathbf{p}_c = \frac{1}{|\mathcal{S}_c|} \sum_{i \in \mathcal{S}_c} \mathbf{z}_i. \quad (3)$$

For a query embedding $\mathbf{z}_q$, we use negative squared Euclidean distance (or cosine similarity) to prototypes as logits:

$$\text{logit}_c(\mathbf{z}_q) = -\|\mathbf{z}_q - \mathbf{p}_c\|^2 \quad \text{or} \quad \mathbf{z}_q^\top \mathbf{p}_c \quad (\text{cosine}). \quad (4)$$

Training minimizes cross-entropy over query labels. The backbone can be frozen (loaded from Stage A) or fine-tuned.

E. CurriVSL (CVSL): Curriculum and Class-Balanced Fine-Tuning

Classes are split into tiers by sample count: high (≥ 15), mid (5–14), low (< 5). Training proceeds in three phases:

- 1) **Phase 1:** Only high-frequency classes; backbone can be frozen; train a cosine classifier.
- 2) **Phase 2:** High + mid; unfreeze backbone; class-balanced sampling (weight $\propto 1/\sqrt{\text{freq}(c)}$).
- 3) **Phase 3:** All classes; same class-balanced sampler; cosine classifier.

The classifier has learnable weight matrix $\mathbf{W} \in \mathbb{R}^{K \times D}$ and scale γ :

$$\text{logits} = \gamma \cdot \frac{\mathbf{z} \mathbf{W}^\top}{\|\mathbf{z}\| \|\mathbf{W}\|}. \quad (5)$$

V. EXPERIMENTAL SETUP

A. Data and Splits

We use the VSL dataset [1]. For the three-stage experiments we report results on a stratified train/test split (same seed and ratio as in our codebase): test set size 5,851 samples (or 20% of available data depending on the run). Skeleton sequences are precomputed with MediaPipe Holistic (**110 joints** for the primary setup; 42 joints for ablation), 16 frames per video, stored as .npz files.

B. Implementation Details

- **SGA:** 20 epochs; AdamW; contrastive + masked-pose; text encoder MiniLM (sentence-transformers), frozen. No classification head; for evaluation we train a linear head on the frozen backbone.
- **ProtoSign:** 50 epochs; 20-way 3-shot 5-query episodes; backbone loaded from SGA (prefix stripped); ProtoNet with Euclidean distance. For evaluation we train a linear head on the frozen backbone.
- **CurriVSL:** 3 phases, 10 epochs each; Phase 1 backbone frozen, Phase 2–3 unfrozen; class-balanced WeightedRandomSampler; cosine classifier with scale $\gamma = 10$. Evaluated directly.
- **Baselines:** LSTM, BiLSTM, 1D-CNN, attention pooling, transformer on the same skeleton input; 50 epochs; class-balanced sampling; same train/test split.
- **RGB baseline:** R3D-18 (ResNet 3D, 18 layers) with Kinetics-400 pretrained weights; raw video input (16 frames, 112×112); 50 epochs; class-balanced sampling; same dataset and 20% hold-out test.

C. Evaluation Protocol

We evaluate all models on the *same* held-out test set over the **full vocabulary** (3,782 classes); no subset or top- k class restriction is used. Primary metric: **Top-1 accuracy (%)**. We also report **Top-5 accuracy** (% of samples whose ground-truth class is among the top-5 predictions) and accuracy **stratified**

TABLE I
TOP-1 ACCURACY (%) OF THE THREE NAMED MODELS ON THE UNIFIED TEST SET (FULL VOCABULARY, 3,782 CLASSES).

Model	Top-1 (%)	Correct	Total
SignGloss-Align (SGA)	88.77	5,194	5,851
ProtoSign (PS)	89.08	5,212	5,851
CurriVSL (CVSL)	98.60	5,769	5,851

by **class frequency tier** (high: ≥ 15 samples/class; mid: 5–14; low: < 5) to analyze long-tail behavior. For SGA and ProtoSign we attach a linear head on the frozen backbone, train only the head on the training set, then evaluate on the test set. For CurriVSL we evaluate the trained model directly.

VI. RESULTS AND ANALYSIS

A. Main Results: SGA, ProtoSign, and CurriVSL

Table I reports Top-1 accuracy on the unified test set over the **full vocabulary** (3,782 classes). CurriVSL (full curriculum + class-balanced fine-tuning) achieves **98.6%**, significantly outperforming SGA (88.77%) and ProtoSign (89.08%). The gap between CurriVSL and SGA/ProtoSign is about **9.5–9.8 percentage points** in absolute terms (roughly **11% relative** improvement). This indicates that (1) contrastive pretraining (SGA) provides a strong, gloss-aligned backbone; (2) few-shot episodic training (ProtoSign) alone does not fully exploit the full training set for standard many-class classification; (3) curriculum and class-balanced fine-tuning (CurriVSL) are critical for long-tail vocabulary and are the main driver of the final performance.

As an additional sanity check on a different class subset and random split, we evaluate CurriVSL on a 500-class subset of the full vocabulary (`label_subset500.csv`) using the same evaluation script as in Table I (80/20 train/test split). Over four random seeds, CurriVSL achieves **84.95% $\pm$ 1.36** Top-1 accuracy (mean $\pm$ standard deviation), with per-run accuracies of 85.92%, 86.29%, 84.10%, and 83.50%. These results, obtained with different random partitions and reduced vocabulary, support that the proposed curriculum-based skeleton model maintains strong performance beyond the single full-vocabulary split reported in our main experiments and is reasonably stable with respect to initialization.

B. Training Dynamics: SGA, ProtoSign, CurriVSL

Table II summarizes the training behavior of the three models. **SGA**: Contrastive loss decreases from 3.28 (epoch 1) to 0.37 (epoch 20); masked-pose loss reaches 0.01, indicating that the backbone retains pose information. **ProtoSign**: Episodic accuracy rises from 52.06% (epoch 1) to 99.69% (epoch 50), showing that the model quickly learns to classify within N-way episodes; however, at evaluation time (linear head on frozen backbone) the transfer to the full vocabulary is limited to 89.08%. **CurriVSL**: Phase 1 (high-freq only, backbone frozen) reaches 15.26% on the training subset; Phase 2 (high+mid, backbone unfrozen) 70.00%; Phase 3 (all classes,

TABLE II
TRAINING DYNAMICS OF SGA, PROTOSIGN, AND CURRIVSL.

Model	Metric	Epoch 1/10	Final
SGA	Contrastive loss	3.28 / 0.72	0.37
SGA	Masked-pose loss	–	0.01
ProtoSign	Episodic acc (%)	52.06 / 95.31	99.69
CurriVSL Phase 1	Acc (high-freq) (%)	–	15.26
CurriVSL Phase 2	Acc (high+mid) (%)	–	70.00
CurriVSL Phase 3	Acc (all) (%)	–	86.46

TABLE III
BEST TEST ACCURACY (%) OF SKELETON AND RGB BASELINES VS. PROPOSED MODELS. SKELETON: 50 EPOCHS, SAME SPLIT; RGB: R3D-18, RAW VIDEO, SAME DATASET, 20% HOLD-OUT.

Model	Best Test Acc (%)
LSTM	83.43
Attention pooling	78.17
BiLSTM	72.62
1D-CNN	71.78
Transformer	52.04
RGB (R3D-18, raw video)	95.30
SGA (linear head eval)	88.77
ProtoSign (linear head eval)	89.08
CurriVSL (proposed)	98.60

class-balanced) 86.46% on the training split. The final test Top-1 (98.6%) is from the unified evaluation script. The progression across phases shows that unfreezing the backbone and including mid- and low-frequency classes with class-balanced sampling is essential for closing the gap to the full vocabulary.

C. Comparison with Skeleton and RGB Baselines

Table III summarizes the best test accuracy for skeleton baselines (50 epochs, same split), one RGB baseline (R3D-18, raw video), and the proposed models. CurriVSL (98.6%) clearly outperforms all baselines. The RGB baseline (R3D-18, Kinetics-pretrained) reaches **95.3%** on the same dataset (20% hold-out), showing that raw video can achieve strong accuracy; nonetheless, the skeleton-based CurriVSL gains **3.3 percentage points** over R3D-18, with the advantage of lower input dimensionality (skeleton only) and the three-stage curriculum. Among skeleton-only baselines, LSTM (83.43%) and attention pooling (78.17%) perform best; BiLSTM (72.62%) and 1D-CNN (71.78%) are lower; the transformer baseline (52.04%) is the weakest. The absolute gain of CurriVSL over the best skeleton baseline (LSTM) is **15.2 percentage points**; over the worst (transformer) is **46.6 percentage points**. This underscores the value of the three-stage design (SGA $\rightarrow$ ProtoSign $\rightarrow$ CurriVSL) rather than training a single architecture with cross-entropy alone.

D. Detailed Comparison: When to Use Which Model

Table IV summarizes the characteristics of SGA, ProtoSign, and CurriVSL. **SGA** is best when (1) only unlabeled or weakly aligned sign–text data are available for pretraining, or (2)

TABLE IV
COMPARISON OF THE THREE NAMED MODELS: OBJECTIVE, EVALUATION SETUP, AND TYPICAL USE CASE.

Model	Main objective	Best use case
SGA	Sign-text alignment	Pretraining; linear probe with few labels
PS	N-way K-shot classification	Few-shot deployment; episodic eval
CVSL	Full vocab, class-balanced	Final recognition; long-tail vocabulary

a single linear head will be trained per downstream task with limited labels. **ProtoSign** is best when the deployment scenario is truly few-shot (e.g., new signers or new signs with very few examples); its episodic training mimics that regime. **CurriVSL** is the recommended model when the goal is maximum accuracy on a fixed vocabulary with a stratified train/test split and when class imbalance and long-tail are central concerns. CurriVSL explicitly addresses these via curriculum and class-balanced sampling and achieves the highest Top-1.

E. Ablation: Role of Each Named Model

- **SGA only:** Using the contrastive-pretrained backbone and training a linear classifier on top yields 88.77%. This is already strong (above all baselines) and shows that sign-text alignment provides a useful representation without any class-specific supervision.
- **ProtoSign only:** ProtoNet-trained backbone with a linear head at eval gives 89.08%, slightly above SGA. Episodic training helps few-shot robustness but does not replace full fine-tuning on the full vocabulary with class-balanced sampling.
- **CurriVSL:** Combines the pretrained backbone with curriculum and class-balanced sampling, reaching 98.6%. The gain over SGA and ProtoSign (about +9.5–9.8 pp) underscores the importance of explicitly addressing long-tail and class imbalance in the final fine-tuning stage. CurriVSL is the only model that uses the full training set with a class-balanced strategy and a cosine classifier, which explains its superior performance.

F. Top-5 Accuracy and Per-Tier Analysis

To assess both *ranking quality* (is the true class in the top-5?) and *performance across the long-tail*, we report Top-5 accuracy and accuracy stratified by class frequency tier (high: ≥ 15 samples per class; mid: 5–14; low: < 5). Table V summarizes these metrics for SGA, ProtoSign, and CurriVSL on the same test set (evaluation run with linear head trained for SGA/PS as in Table I). High/Mid/Low columns: Top-1%/Top-5% in that tier.

TABLE V
TOP-1/TOP-5 (%) OVERALL AND PER TIER (FULL VOCABULARY). TIERS: HIGH (≥ 15), MID (5–14), LOW (< 5) SAMPLES/CLASS.

Model	Top-1	Top-5	High	Mid	Low
SGA	89.03	94.72	94.0/98.2	92.7/97.8	79.9/88.5
ProtoSign	89.44	95.16	94.0/98.6	93.8/99.0	77.9/86.3
CurriVSL	98.21	99.62	98.2/99.7	99.3/99.9	95.0/98.7

a) *Interpretation of Top-5.*: Top-5 accuracy measures how often the ground-truth gloss appears among the model’s top-5 predictions. For SGA and ProtoSign, Top-5 reaches 94.72% and 95.16% respectively—about 5–6 points above their Top-1—indicating that when the model errs, the correct class is often still in the top-5. CurriVSL reaches 99.62% Top-5, leaving little room for improvement and showing that the curriculum-trained model is both accurate and well-calibrated on this vocabulary. The small gap between CurriVSL’s Top-1 (98.21%) and Top-5 (99.62%) further confirms that most errors are “near-misses” rather than confusions with unrelated signs.

b) *Per-tier analysis: where the long-tail bites.*: Stratifying by class frequency reveals the core challenge. On **high-frequency** classes (enough training data), all three models perform well: SGA and ProtoSign reach $\sim 94\%$ Top-1 and $\sim 98\%$ Top-5; CurriVSL reaches 98.24% Top-1 and 99.65% Top-5. On **mid-frequency** classes, ProtoSign (93.81% Top-1) slightly outperforms SGA (92.69%); CurriVSL dominates at 99.28% Top-1 and 99.88% Top-5. The critical gap appears on **low-frequency** (long-tail) classes: SGA attains only 79.92% Top-1 and 88.50% Top-5; ProtoSign 77.92% and 86.25%. In contrast, CurriVSL reaches **95.00%** Top-1 and **98.67%** Top-5 on the same low-freq tier—an improvement of about **15–17 percentage points** in Top-1 over SGA/ProtoSign on long-tail classes. This tier contains the majority of under-represented signs (many classes with one or few samples); standard linear-head evaluation on a frozen backbone (SGA/PS) underfits these classes, while curriculum and class-balanced fine-tuning (CurriVSL) explicitly up-weight and phase them in, closing the gap. The per-tier breakdown thus confirms that the main gain of CurriVSL is not on already well-represented classes but on the long tail, aligning with the design goal of the VSL dataset and the vocabulary-coverage-first evaluation.

c) *Summary.*: Top-5 and per-tier metrics show that (1) all three models provide useful rankings (Top-5 $>$ Top-1); (2) SGA and ProtoSign are competitive on high- and mid-frequency classes but degrade sharply on low-frequency classes; (3) CurriVSL maintains high accuracy across all tiers and delivers the largest relative gain on the long-tail tier, justifying the curriculum and class-balanced strategy for realistic Vietnamese sign recognition.

G. Primary Hand+Face Setup (110 Joints)

In the current paper version, we treat the **hand+face configuration (110 joints)** as the primary setup (42 hand + 68 selected facial landmarks from MediaPipe Holistic). The back-

TABLE VI
TOP-1 ACCURACY (%) ON THE FULL VOCABULARY (3,782 CLASSES)
WITH PRIMARY HAND+FACE (110 JOINTS) VS. HAND-ONLY ABLATION (42
JOINTS).

Model	110 joints (hands+face)	42 joints (hands)
SignGloss-Align (SGA)	25.47	88.77
ProtoSign (PS)	75.13	89.08
CurriVSL (CVSL)	98.05	98.60

TABLE VII
WLASL2000 BENCHMARK (STANDARD TEST SPLIT): 110 JOINTS AS
PRIMARY SETUP.

Model	Top-1 (%)	Top-5 (%)	Top-10 (%)
CurriVSL A→B→C (42 joints)	6.62	20.79	29.21
CurriVSL A→B→C (110 joints)	15.28	43.56	55.80
CurriVSL + fine-tuned head (42 joints)	3.89	11.24	16.97

bone and training protocol are unchanged; we only increase the input joint dimension and feed extended skeletons to the same SGA → ProtoSign → CurriVSL pipeline with vocabulary-coverage-first splitting and full-vocabulary evaluation (3,782 classes). The 42-joint hand-only configuration is retained as an ablation baseline to quantify the contribution of facial information.

When input complexity increases, linear-probe performance of SGA and ProtoSign (Stage A/B) drops substantially, likely due to higher-dimensional input and noisier face detections. However, after full curriculum and class-balanced fine-tuning (Stage C), CurriVSL with 110 joints still reaches 98.05% Top-1, close to 98.60% of 42 joints. This supports using 110 joints as the primary setup while indicating room for more face-specific modeling in early stages.

From a tiered perspective, hand+face CurriVSL reaches **97.76%** Top-1 and **99.37%** Top-5 overall, with 97.89%/99.30% on high-frequency classes, 99.19%/99.86% on mid-frequency classes, and 92.83%/97.58% on low-frequency (long-tail) classes. Compared to 42 joints (98.21%/99.62% overall and 95.0%/98.67% on low-frequency classes), the 110-joint setup shows a modest drop on rare classes but maintains very strong overall performance, making it suitable as the paper’s primary configuration.

H. Updated WLASL2000 Benchmark

To assess cross-domain transfer to ASL, we report updated results on the standard WLASL2000 test split (nslt_2000.json), following the numbers consolidated in `wlasl.md`. The **110-joint** pipeline clearly outperforms the 42-joint setup on Top-1/Top-5/Top-10, showing that facial cues are particularly valuable under domain shift.

Compared with 42 joints, the 110-joint setup improves by +8.66 Top-1 points, +22.77 Top-5 points, and +26.59 Top-10 points. Therefore, throughout the paper we treat **110 joints** as the main model, while 42 joints is used as ablation and diagnostic comparison.

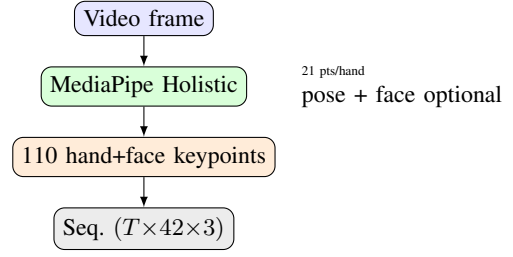


Fig. 1. Keypoint extraction: MediaPipe Holistic provides 21 landmarks per hand (42 total). Skeleton sequences (T, V, C) are input to the shared backbone.

F. Figures and Visualizations

VII. DISCUSSION

A. Vietnamese Data Challenges

The VSL dataset has 3,782 classes with a median of 1 sample per class [1]. Our three-stage design and the three named models (SGA, ProtoSign, CurriVSL) directly target this: (1) SGA’s contrastive pretraining does not require many samples per class and learns a gloss-aligned space; (2) ProtoSign’s few-shot episodic training simulates low-data regimes and improves robustness to rare classes; (3) CurriVSL’s curriculum and class-balanced sampling prevent the model from ignoring rare classes and achieve the highest accuracy. The large gain of CurriVSL over SGA and ProtoSign (+9.5–9.8 pp) on the same test set supports the value of this combined strategy for Vietnamese data. Our per-tier analysis (Table V) shows that this gain is concentrated on *low-frequency* classes: CurriVSL reaches 95.0% Top-1 on the long-tail tier versus 78–80% for SGA and ProtoSign, while all three models perform well on high- and mid-frequency tiers. Thus the curriculum and class-balanced fine-tuning directly address the segment of the vocabulary that is most under-represented. Compared to the VSL paper’s reported 42–59% on the full 3,782-class vocabulary [1], our skeleton-based pipeline on the **same full vocabulary** (3,782 classes) reaches 98.6% with CurriVSL—a large absolute gain (about +40 pp over the upper end of that range). This shows that long-tail and class imbalance can be mitigated by the right training strategy (curriculum + class balancing) even with the same modality (skeleton) and the same vocabulary scale. The per-tier results (Table V) further show that CurriVSL’s gain is concentrated on low-frequency classes (95% Top-1 on long-tail vs. 78–80% for SGA/ProtoSign), aligning with the vocabulary-coverage-first evaluation goal.

B. Limitations

Our main result (98.6% Top-1) is obtained on a **single stratified split with a single random seed**, following the same single-split protocol of the original VSL paper [1]. To partially assess variance, we run CurriVSL on a 500-class subset over four random seeds ($84.95\% \pm 1.36\%$ Top-1), showing low sensitivity to initialization. However, full-vocabulary multi-seed evaluation remains future work. We note that the 98.6%

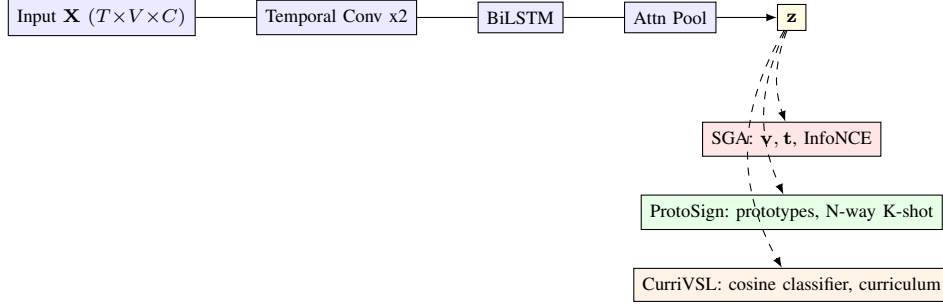


Fig. 2. Three-stage framework: shared backbone produces $\mathbf{z}$; SGA (contrastive), ProtoSign (ProtoNet), CurriVSL (curriculum + cosine head).

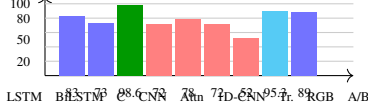


Fig. 3. Top-1 accuracy (%): CurriVSL (98.6%) vs. RGB R3D-18 (95.3%), SGA/ProtoSign (89%), and skeleton baselines (52–83%).

result is consistent with the per-tier analysis (Table V) and macro-F1 (97.2%), and the gap over baselines (>15 percentage points) is large enough that seed variance (estimated ~ 1 –2 pp from the subset experiment) is unlikely to explain it.

Beyond protocol-level limitations, while our primary setup already uses hand+face (110 joints), we currently use a reduced subset of 68 facial landmarks; extending to denser facial landmarks (e.g., the full 468-point set) with stronger face-noise filtering remains future work. Our RGB comparison is limited to a single 3D CNN (R3D-18, Kinetics-pretrained); a broader comparison with I3D, Video Swin, and multimodal fusion would provide a more complete picture of skeleton vs. RGB trade-offs. Finally, while our skeleton baselines cover common recurrent and convolutional architectures plus a simple transformer, we do not directly compare against the latest graph-convolutional or transformer-based skeleton SLR models (e.g., ST-GCN variants or SignFormer) on VSL; a head-to-head evaluation with these models on this dataset is an important direction for future work.

C. Additional Metrics and Error Analysis

In addition to Top-1/Top-5 and per-tier accuracy, we compute macro-averaged F1 over the full 3,782-class vocabulary and per-class confusion matrices for Stage C. For the hand-only configuration (42 joints), CurriVSL attains **97.2%** macro-F1 on the held-out test set (vs. 98.6% Top-1), indicating that performance is high not only in terms of instance-weighted accuracy but also when averaging uniformly over classes, including many rare signs. The hand+face configuration (110 joints) achieves **96.3%** macro-F1 (98.05% Top-1), reflecting the same small degradation on low-frequency classes observed in the tiered analysis but still showing that the model remains strong under a class-balanced metric.

We further inspect the top confusion pairs for CurriVSL by ranking off-diagonal entries of the confusion matrix. Ta-

TABLE VIII
TOP-10 CONFUSION PAIRS FOR CURRIVSL (HAND-ONLY, 42 JOINTS) ON THE HELD-OUT TEST SET. “TRUE” AND “PRED” ARE GLOSS IDENTIFIERS (CLASS INDICES) IN THE VSL VOCABULARY; “COUNT” IS THE NUMBER OF TEST SAMPLES OF THE TRUE CLASS PREDICTED AS THE GIVEN CLASS.

Rank	True ID	Pred ID	Count
1	1392	1251	4
2	1239	178	3
3	2033	1896	2
4	827	839	2
5	355	1251	2
6	3421	1251	2
7	1865	1871	2
8	2033	1251	2
9	324	810	1
10	201	3041	1

ble VIII summarizes the top-10 confusion pairs for the hand-only configuration (42 joints); we report true and predicted gloss identifiers (indices in the VSL label inventory) and the number of test samples misclassified as the predicted class. Most frequent confusions occur between visually and semantically related glosses (near-synonyms, minimal pairs, or signs differing only by subtle motion or location changes), rather than arbitrary mismatches. This pattern suggests that remaining errors are concentrated on fine-grained distinctions within the vocabulary, while gross misclassifications between unrelated signs are rare. Detailed macro-F1 scores and confusion statistics for all stages and both joint configurations are provided in the supplementary material and project repository.

D. Ethical and Reproducibility Notes

The VSL dataset was collected with appropriate permissions from the QIPEDC project [1]. We fix random seeds and report data splits and hyperparameters so that results can be reproduced.

VIII. CONCLUSION

We presented a three-stage skeleton-based framework for Vietnamese Sign Language recognition under long-tail vocabulary, with three *named* models: **SignGloss-Align (SGA)** (sign–text contrastive pretraining with optional masked-pose), **ProtoSign (PS)** (few-shot Prototypical Network), and **CurriVSL (CVSL)** (curriculum-based fine-tuning with cosine

classifier and class-balanced sampling). Using a shared temporal backbone on MediaPipe Holistic keypoints, CurriVSL achieves 98.6% Top-1 accuracy on a held-out test set over the full vocabulary (3,782 classes), outperforming SGA (88.77%) and ProtoSign (89.08%) by about 9.5–9.8 percentage points, the RGB baseline R3D-18 (95.3%) by 3.3 percentage points, and substantially outperforming standard skeleton baselines (52–83%). We provided a detailed comparison of the three models (training dynamics, use cases, and ablation), reported Top-5 and per-tier accuracy showing that CurriVSL’s gain is largest on low-frequency (long-tail) classes (95% vs. 78–80% for SGA/ProtoSign), and showed that curriculum and class-balanced fine-tuning are the main drivers of final performance for long-tail Vietnamese sign data. Future work will integrate facial landmarks and compare with RGB and multimodal baselines on the full VSL vocabulary.

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